$$A = e^{A_S T} \qquad b = \int_0^T e^{A_S \nu} b_S \mathrm{d}\nu$$

$$G(s) = c[sI - A_S]^{-1} b_S + d$$

$$\begin{aligned} \dot{x}(t) &= A_S x(t) + b_S u(t) \\ y(t) &= c x(t) + d u(t) \end{aligned}$$

$$\begin{aligned} x(k+1) &= A x(k) + b u(k) \\ y(k) &= c x(k) + d u(k) \end{aligned}$$

$$\bar{G}(z) = c[zI - A]^{-1} b + d$$

zeitkontinuierlich

zeitdiskret

ZRD

UTF

$$Y(s) = G(s) U(s)$$

$$Y(z) = \bar{G}(z) U(z)$$

$$\bar{G}(z) = \left(\frac{z-1}{z}\right) \mathcal{Z}\left\{ \left. \left(\mathcal{L}^{-1}\left\{ \frac{G(s)}{s} \right\} \right) \right|_{t=kT} \right\}$$